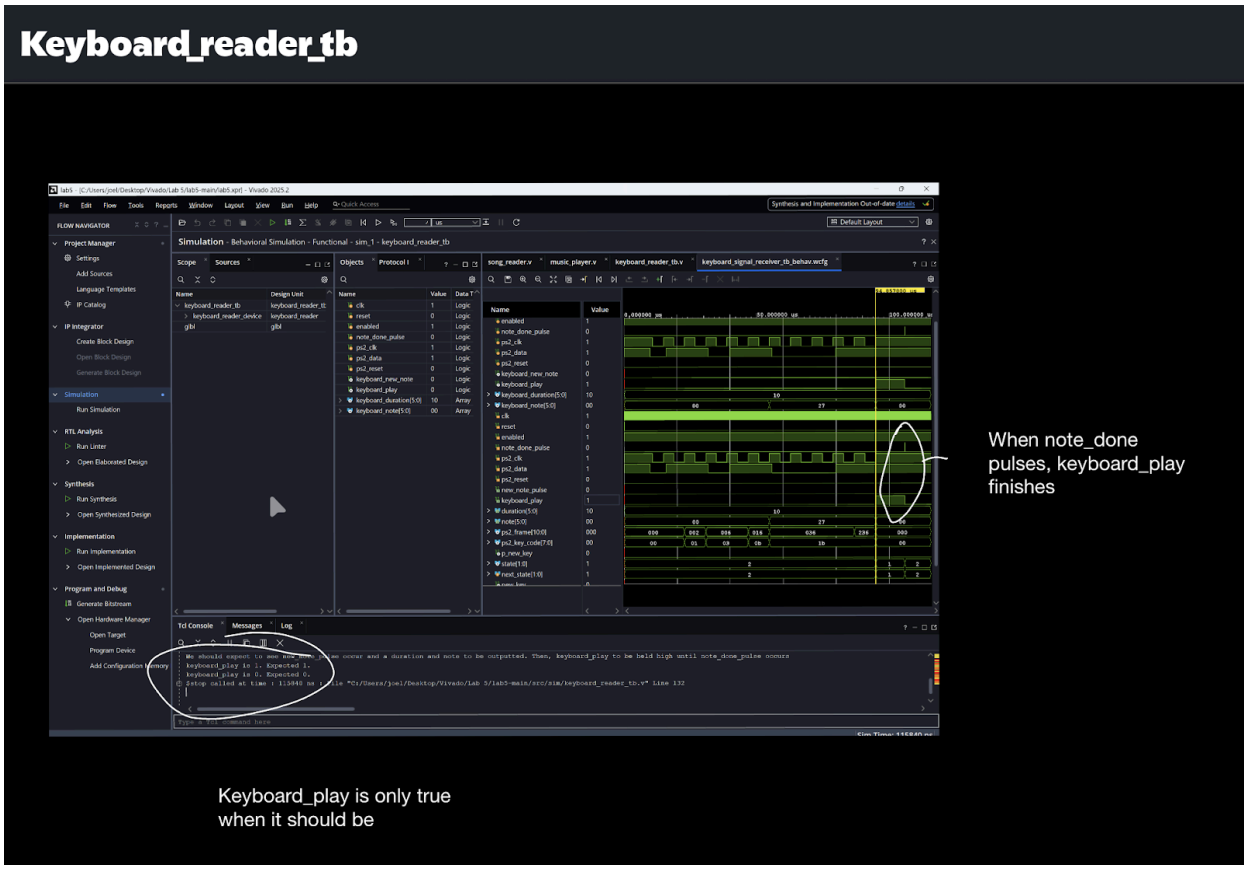
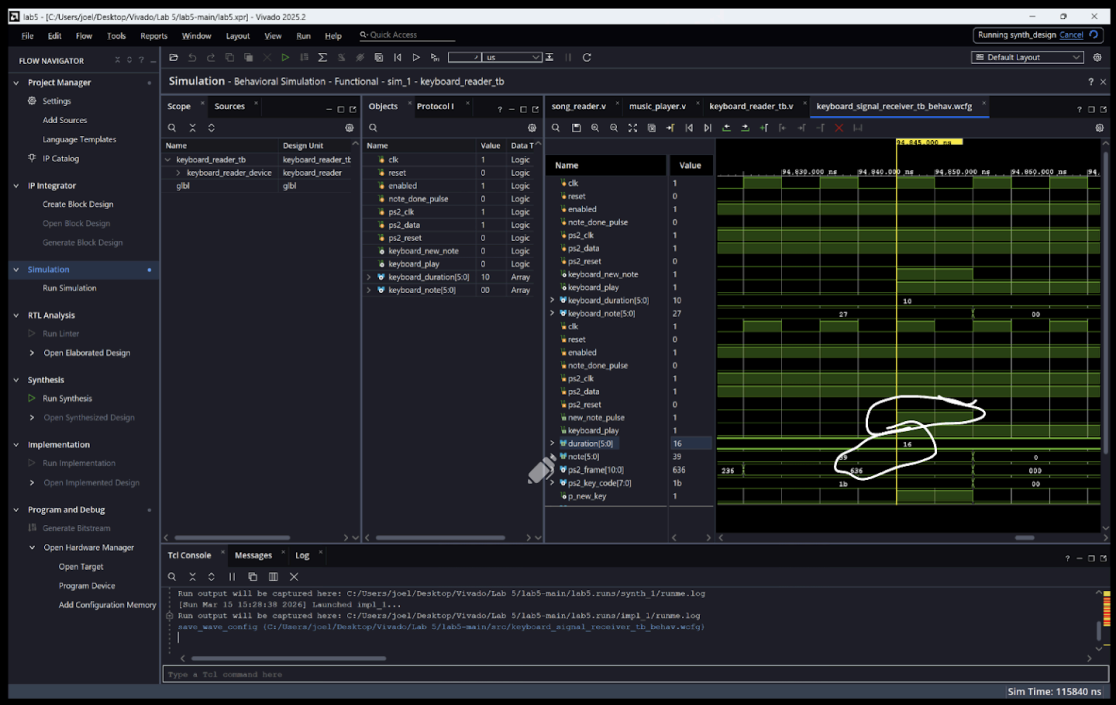
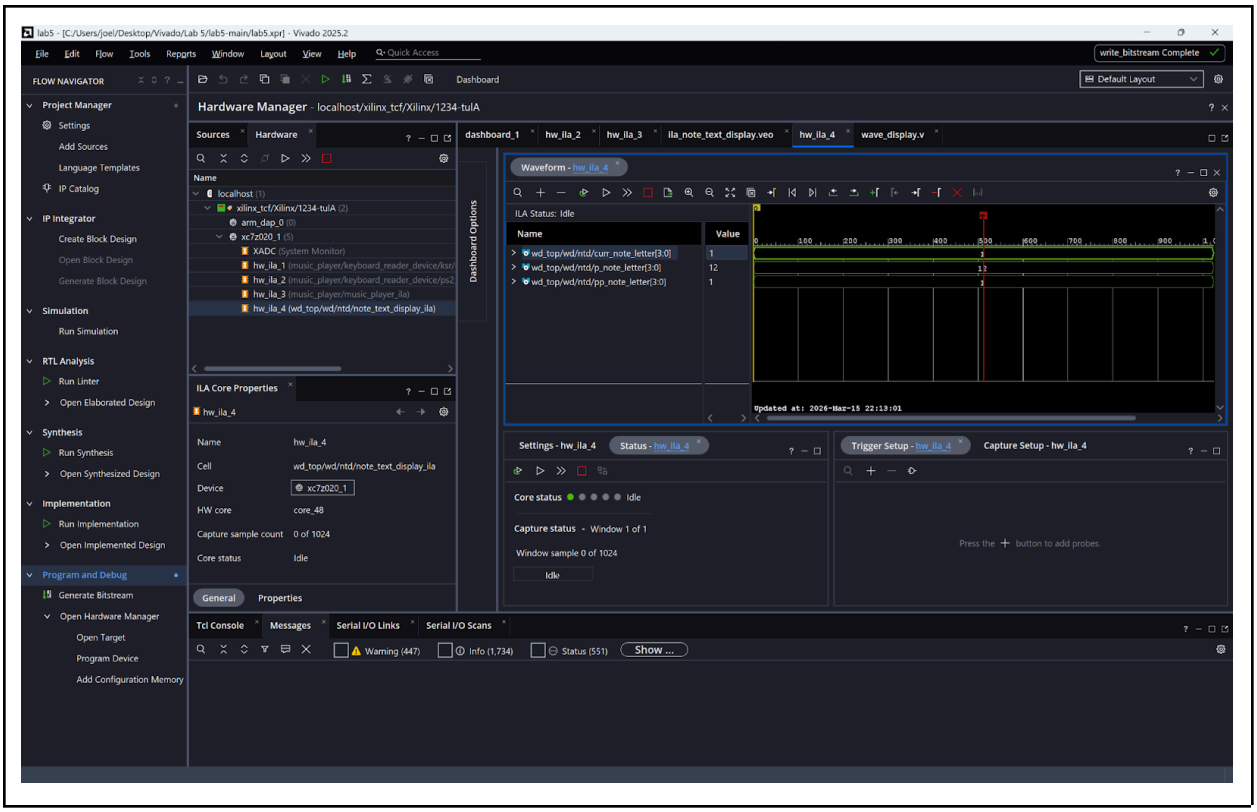
Keyboard_reader_tb



Zooming in a bit, we get the waveform image below. We can see that during the new_note_pulse, the duration and note are the right values (39 for note and the duration is constantly 16). We also see that the keyboard_play is 1 during the new_note_pulse.



We can see in this ILA that curr_note_letter, p_note_letter, and pp_note_letter are all being set on the FPGA (corresponds to A, G#, and A)


lab5 - [C:/Users/joel/Desktop/Vivado/Lab 5/lab5-main/lab5.xpr] - Vivado 2025.2
File Edit Flow Tools Reports Window Layout View Help
write_bitstream Complete
FLOW NAVIGATOR
Project Manager
Settings
Add Sources
Language Templates
IP Catalog
IP Integrator
Create Block Design
Open Block Design
Generate Block Design
Simulation
Run Simulation
RTL Analysis
Run Linter
Open Elaborated Design
Synthesis
Run Synthesis
Open Synthesized Design
Implementation
Run Implementation
Open Implemented Design
Program and Debug
Generate Bitstream
Open Hardware Manager
Open Target
Program Device
Add Configuration Memory
Dashboard
Default Layout
Hardware Manager - localhost/xilinx_tcf/Xilinx/1234-tulA
Sources
Hardware
localhost (1)
xilinx_tcf/Xilinx/1234-tulA (2)
arm_dap_0 (0)
xc7z020_1 (5)
XADC (System Monitor)
hw_ila_1 (music_player/keyboard_reader_device/kst/
hw_ila_2 (music_player/keyboard_reader_device/ps2
hw_ila_3 (music_player/music_player_ila)
hw_ila_4 (wd_top/wd/ntd/note_text_display_ila)
ILA Core Properties
Name hw_ila_4
Cell wd_top/wd/ntd/note_text_display_ila
Device xc7z020_1
HW core core_48
Capture sample count 0 of 1024
Core status Idle
General
Properties
Dashboard Options
dashboard_1
hw_ila_2
hw_ila_3
ila_note_text_display.veo
hw_ila_4
wave_display.v
Waveform - hw_ila_4
ILA Status: Idle
Name
Value
wd_top/wd/ntd/curr_note_letter[3:0] 1
wd_top/wd/ntd/p_note_letter[3:0] 12
wd_top/wd/ntd/pp_note_letter[3:0] 1
Updated at: 2026-Mar-15 22:13:01
Settings - hw_ila_4
Status - hw_ila_4
Core status Idle
Capture status - Window 1 of 1
Window sample 0 of 1024
Idle
Trigger Setup - hw_ila_4
Capture Setup - hw_ila_4
Press the + button to add probes.
Tcl Console
Messages
Serial I/O Links
Serial I/O Scans
Warning (447)
Info (1,734)
Status (551)
Show ...